

ARTIFICIAL INTELLIGENCE, ROBOTICS AND BIG DATA ANALYTICS

These technologies are transforming the modern economy and society by making systems smarter, faster, and more efficient.

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INTRODUCTION

Artificial Intelligence, Robotics, and Big Data are modern technologies that play an important role in development. They are widely used in industries, agriculture, healthcare, education, and governance. Together, they support sustainable and digital growth.

WHAT IS ARTIFICIAL INTELLIGENCE (AI)?

Artificial Intelligence refers to machines that can think, learn, and make decisions like humans. AI systems use data and algorithms to perform tasks such as recognizing speech, images, and patterns.



FEATURES OF AI

AI has the ability to learn from experience and improve over time. It can analyze large amounts of data quickly and make accurate decisions with minimal human intervention.



TYPES OF AI

- **Narrow AI** performs specific tasks like voice assistants.
- **General AI** aims to perform all intellectual tasks like humans.
- **Super AI** refers to machines that surpass human intelligence, which is still theoretical.

USES OF AI

AI is used in healthcare for disease diagnosis

AI is used in agriculture for crop monitoring,

AI is used in banking for fraud detection, and in governance for efficient public service delivery

ADVANTAGES OF AI

AI reduces human error, saves time, and increases efficiency. It works continuously without fatigue and helps in handling complex and repetitive tasks.

LIMITATIONS OF AI

AI systems are costly to develop and maintain. Over-dependence on AI may lead to job losses and ethical issues related to data privacy and security.

WHAT IS ROBOTICS?

Robotics is a branch of technology that deals with the design and use of robots. Robots are machines capable of performing tasks automatically with high precision.

TYPES OF ROBOTS

Different robots are designed for different purposes, such as industrial robots for factories, medical robots for surgeries, and service robots for household and public use.

USES OF ROBOTICS

Robots are used in manufacturing to increase production, in healthcare for accurate surgeries, and in space exploration where human presence is risky.

ADVANTAGES OF ROBOTICS

Robots increase productivity, reduce human risk, and ensure consistent quality. They are useful in hazardous environments where human safety is a concern.

WHAT IS BIG DATA ANALYTICS?

Big Data Analytics involves examining large and complex data sets to discover patterns and useful information that help in better decision-making.

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FEATURES OF BIG DATA (5VS)

Big Data is characterized by huge volume, fast speed of generation, different types of data, accuracy of data, and its usefulness in decision-making.

USES OF BIG DATA ANALYTICS

Big Data helps businesses predict consumer behavior, governments frame effective policies, and healthcare systems improve treatment and planning.





RELATIONSHIP AMONG AI, ROBOTICS & BIG DATA

Big Data provides information, AI analyzes that data, and robotics applies AI to perform tasks. Together, they form intelligent and automated systems.



IMPACT ON ECONOMY

These technologies increase productivity, reduce costs, promote innovation, and generate new employment opportunities in skilled sectors.



CONCLUSION

Artificial Intelligence, Robotics, and Big Data Analytics are essential for modern development. With proper regulation and skill development, they can ensure sustainable and inclusive growth.



Q1. ROLE OF AI AND ROBOTICS IN MODERN ERA

Artificial Intelligence and robotics play an important role in the modern era by transforming industries and daily life. They help automate repetitive and dangerous tasks, improve accuracy in healthcare, support personalized education, and enable smart transportation systems. AI-powered machines increase efficiency, reduce human effort, and enhance decision-making. When used responsibly, AI and robotics contribute to economic growth, technological advancement, and a better quality of life for society.



Q2. APPROACHES USED BY SCIENTISTS TO ENSURE ACCURACY AND RELIABILITY OF SCIENTIFIC DATA WITH AI AND ROBOTICS

Scientists ensure accuracy and reliability of scientific data with AI and robotics by carefully validating and cleaning data, and regularly calibrating sensors and instruments. AI models are trained and tested on high-quality datasets to reduce errors and bias. Human experts review and verify AI results to avoid mistakes. Experiments are repeated and cross-checked for consistency. Proper documentation and transparency of methods also help maintain trust and reliability in scientific research.

